

# Mid-Semester Presentation:

# Self-adjusting Photolithography Lens



**Presented By:**

- ❑ M i c h a e l    H o r o w i t z
- ❑ M i c h a e l    M o s k o w i t z
- ❑ J a c o b        G o l d s m i t h
- ❑ J o s e p h      P i c a r d
- ❑ J a c o b        R a b i n o w i t z

**Sponsor rep:**

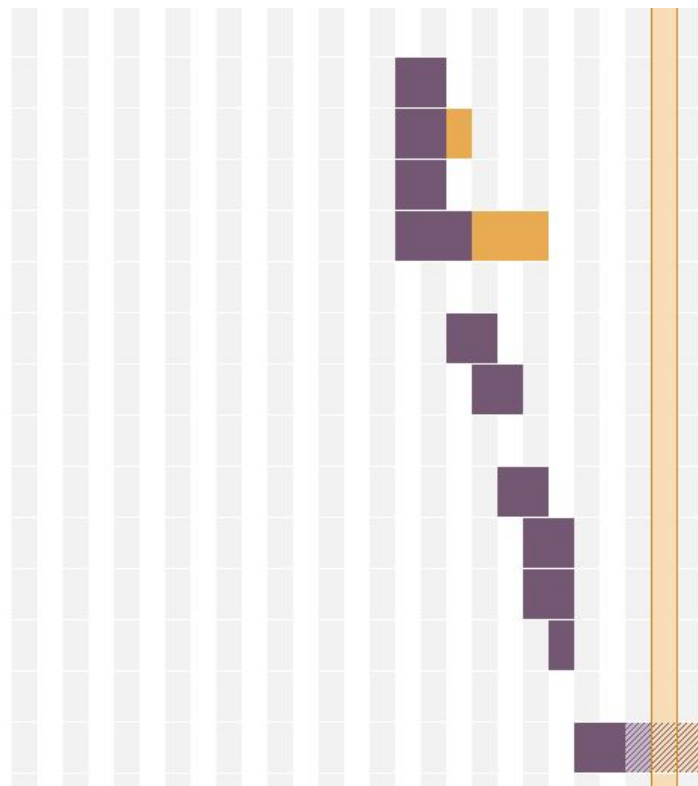
**Mathew Sorna, ASML**

# Contents

- **Project Progress**
- **Project Description**
- **Previous semester's work**
- **Component Purchase**
- **Component Redesign**
- **Next Steps**

# Gantt Chart

| Orders placed                                       |    |   |    |   |      |
|-----------------------------------------------------|----|---|----|---|------|
| Ordered Ball and Socket joints                      | 17 | 2 | 17 | 2 | 100% |
| Ordered Lens                                        | 17 | 2 | 17 | 3 | 100% |
| Ordered beams for test stand                        | 17 | 2 | 17 | 2 | 100% |
| Found Alternative Actuators and Ordered Them        | 17 | 3 | 17 | 6 | 100% |
| Redesign of Mounting and Flextures                  |    |   |    |   |      |
| Redesign of Flextures                               | 19 | 2 | 19 | 2 | 100% |
| Cut mounting ring in half for ease of manufacturing | 20 | 2 | 20 | 2 | 100% |
| Designing of New Components and 3D model build      |    |   |    |   |      |
| Designed Place for base to sit in T-slots           | 21 | 2 | 21 | 2 | 100% |
| Designed Motor Holders                              | 22 | 2 | 22 | 2 | 100% |
| Designed Base for System                            | 22 | 2 | 22 | 2 | 100% |
| Built a 3D model of design                          | 23 | 1 | 23 | 1 | 100% |
| Testing                                             |    |   |    |   |      |
| Testing of motors                                   | 24 | 3 | 24 | 5 | 50%  |



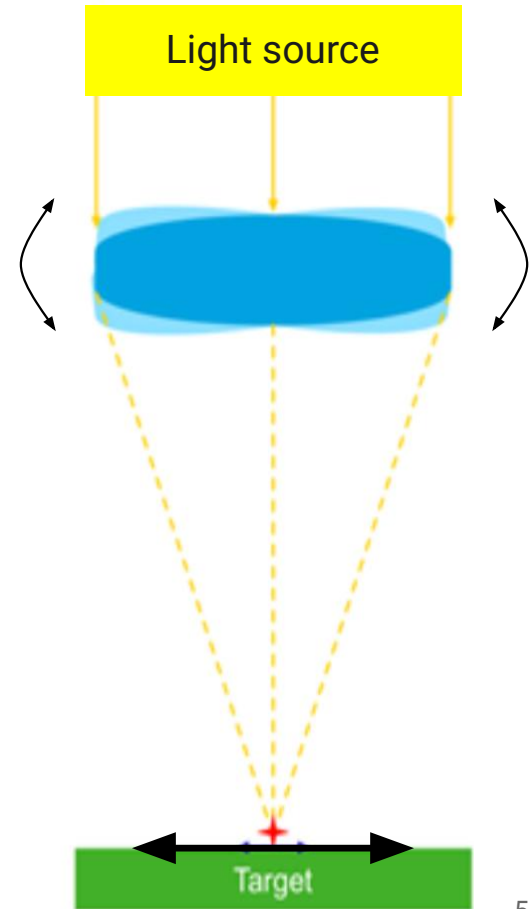
# The precise control of light

(Background and motivation)

- 
- A diagram illustrating the precise control of light. It features a dark blue, upward-sloping surface on the right side. Three yellow lines with arrowheads at their ends originate from a point on this surface and extend upwards and to the left. Three other yellow lines with arrowheads at their ends originate from a point further to the left and extend downwards and to the right, converging towards the surface. This visualizes the reflection and focusing of light rays.
- In ASML's EUV Photolithography machine, an abundance of mirrors and lenses are implemented to precisely control the direction and use of light, including high energy ultraviolet light
  - Light is a tool used in many industries, from medical imaging and radiation treatments to consumer cameras
  - It is therefore important to find methods to control and direct light rays

# Project Description/ Problem statement

- We must design a mechanical system to tilt an optic lens along two axes, to shift a focal point.
- The system should be self-adjusting to account for vibration and thermal disturbances.
- Have a radially compliant mounting-mechanism to reduce any stresses in the lens induced from thermal expansion.



# Requirements and Specifications

## Requirements

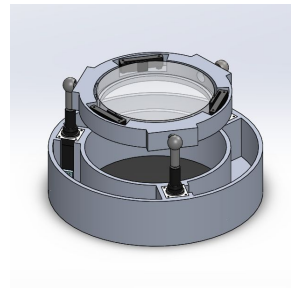
- Lens-tilting capacity: in the range of +/- 10 degrees in the Rx and Ry directions
- Movements in-plane: should have a 1000:1 ratio to lateral movements between planes

## Design Parameters

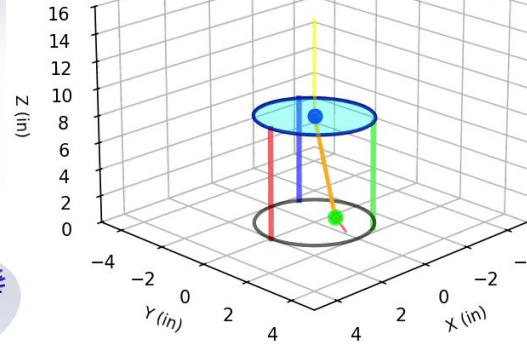
- Lens optic: is specified between 4-8" Outer Diameter
- Mounting ring: between a 6" Inner diameter and a 10" OD
- Lens positioning: ideally occur in under 3 seconds

# Summary of Previous semester's work

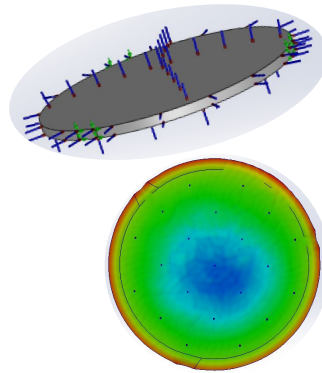
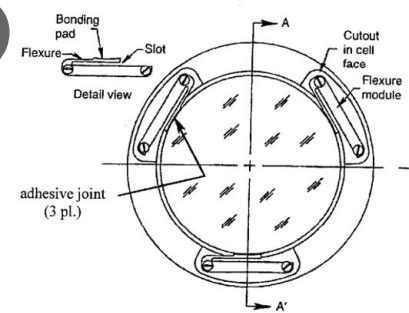
1. Finalized design concept
2. Thorough literature search
3. Performed FEA analysis on the lens and flexures
4. Selected our lens, motor, and other components
5. Presented a layout of the system software
6. 3D Printed and assembled a full mockup of the design



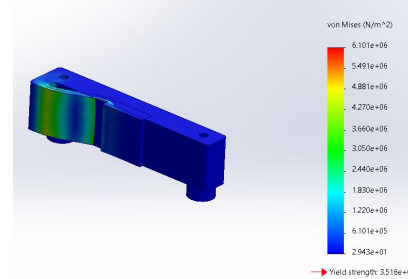
(1)



(2)



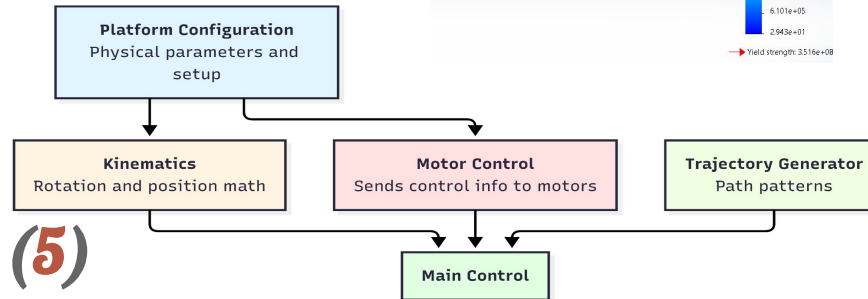
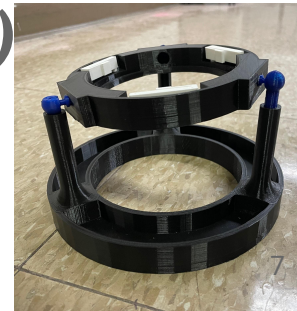
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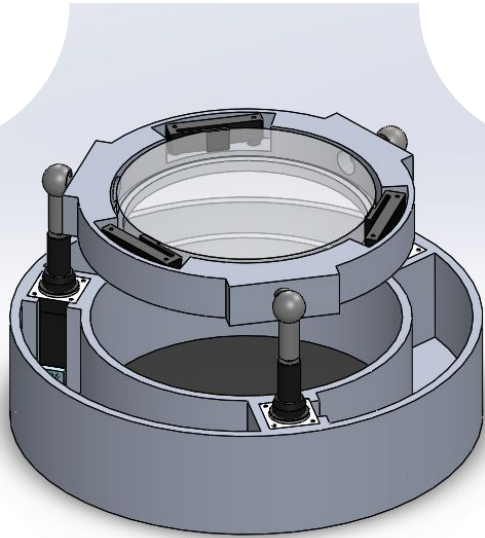
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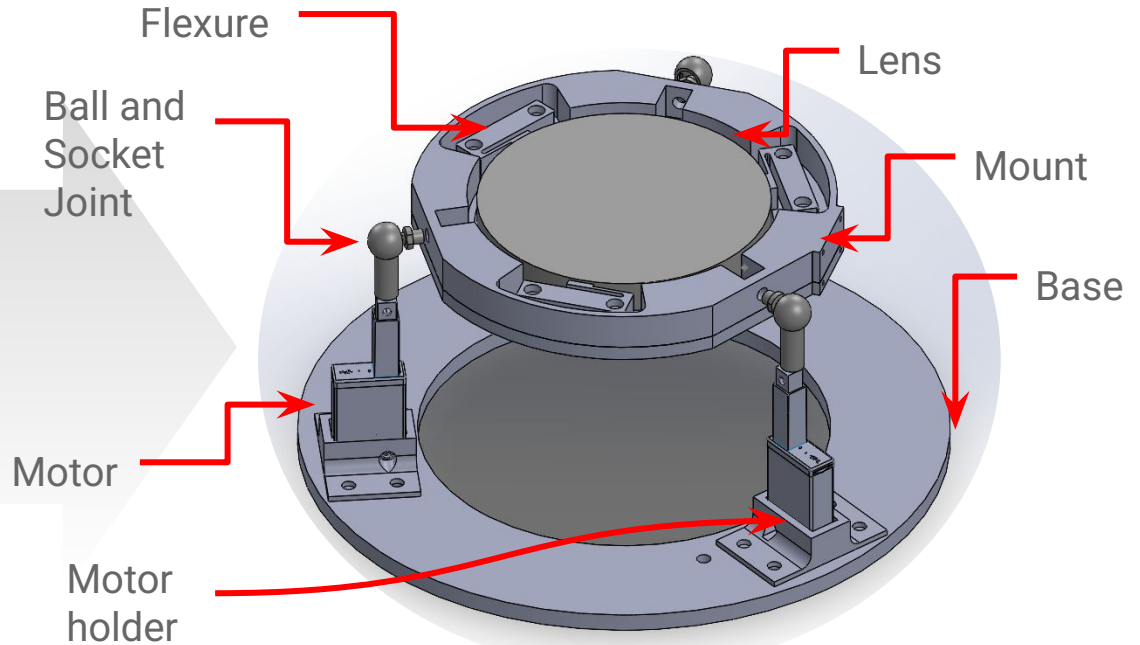
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# Labeled View of Linear Actuator Model

(end of semester design)



(Previous semester's design)



(Improved design)



# Bill of Materials

| PART NUMBER           | PART NAME                                                                                             | DESCRIPTION                                                                             | QUANTITY | UNITS | PART IMAGE                                                                          | UNIT COST | AMOUNT    |
|-----------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------|-------|-------------------------------------------------------------------------------------|-----------|-----------|
| 2RCP8                 | T-Slot Framing Extrusion: 10 Series, 6 ft Lg, 1 in x 1 in, Silver, Single, Grooved, 4 Slots           |                                                                                         | 2        | 1     |  | \$ 41.75  | \$ 83.50  |
| 856NE0                | JW WINCO Ball Joint Linkages: Male Studded, Right Hand Thread, Alloy Steel, 12.8 mm Head Outside Dia  |                                                                                         | 4        | 1     |  | \$ 8.27   | \$ 33.08  |
| 2495-P8-25-49-3-ST-ND | P8-25-49-3-ST                                                                                         | Linear Actuator Stepper, Bipolar 0.984" (25.00mm) Ball Screw Includes Controller 4.2VDC | 3        | 1     |  | \$ 90.00  | \$ 270.00 |
| 3886                  | 1528-3886-ND                                                                                          | STEMMA QT MPU-6050 ACCEL GYRO                                                           | 1        | 1     |  | \$ 12.95  | \$ 12.95  |
| 2183-2134-ND          | 2134                                                                                                  | DRV8834 STEPPER MTR DRVR CARRIER                                                        | 4        | 1     |  | \$ 9.95   | \$ 39.80  |
| 1050-1018-ND          | A000067                                                                                               | ARDUINO MEGA2560 ATMEGA2560                                                             | 1        | 1     |  | \$ 49.90  | \$ 49.90  |
| 8975K222              | Multipurpose 6061 Aluminum Bar                                                                        | 1/2" Thick x 8" Wide                                                                    | 1        | 1     |  | \$ 57.29  | \$ 57.29  |
|                       | Uxcell 300mm Focal Length Double Convex Lens, 2 Pack 100mm Diameter 11mm Height Glass Optically Lens, |                                                                                         | 1        | 2     |  | \$ 18.59  | \$ 18.59  |
| TOTAL PARTS           |                                                                                                       |                                                                                         | 17       | TOTAL |                                                                                     | \$        | 565.11    |

# Linear actuator: Reselection and purchase

## Original NEMA 8 Captive Linear Stepper Motor

- 19.1mm stroke
- Couldn't be ordered due to shipping troubles



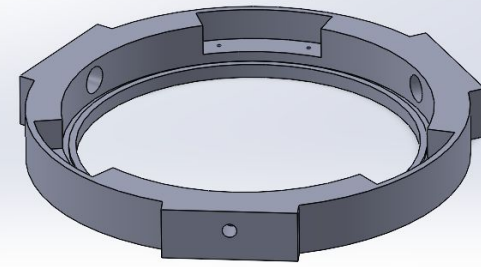
## P8-ST Micro Stepper Actuator

- 25mm stroke
- Low voltage requirement
- Can still handle our loads and requirements
- Cheaper and faster shipping

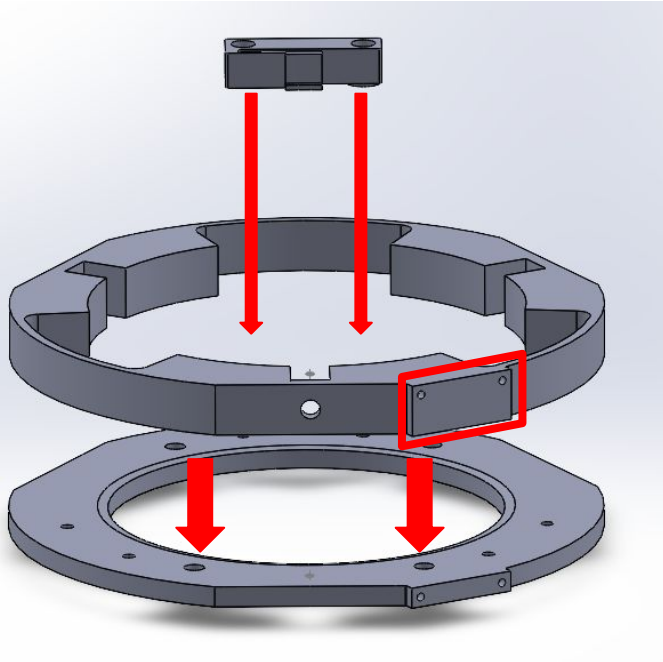


**The new Actuator presented its own unique assembly problems?**

# Mounting Ring Final Design design



(Old Design)

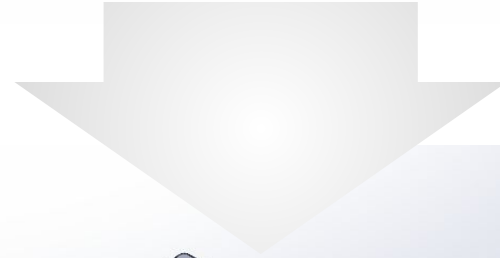
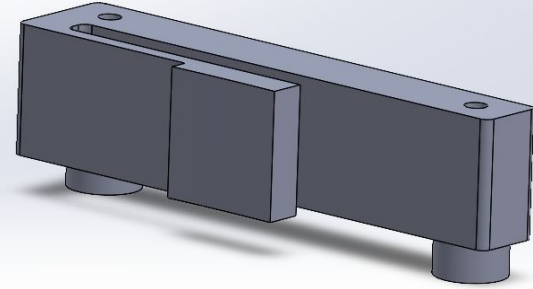


- Split into two sections for machinability
- Enlarged screw holes for easier assembly (plates and ball and socket joints)
- Enlarged entire fixture to account for enlarge screws.
- Connection Plane added on side of ring to accommodate the accelerometer

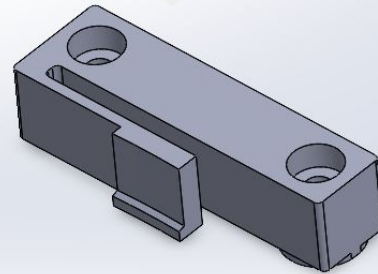
# Flexures Final Design

- Enlarged body to accommodate bigger connecting screws
- Lip at bottom of connecting face to ensure the lens isn't resting on mounting ring
- The legs are replaced with washers to allow for more easy machinability.

Old  
Flexure



New  
Flexure



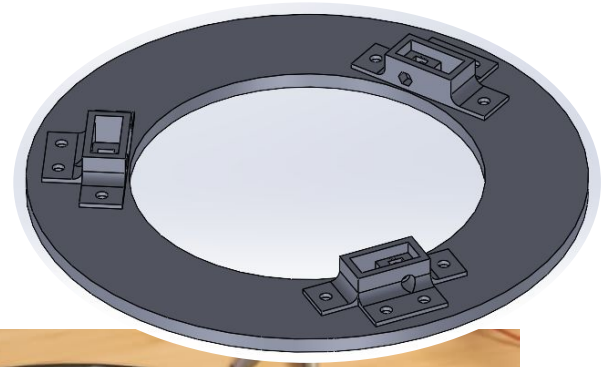
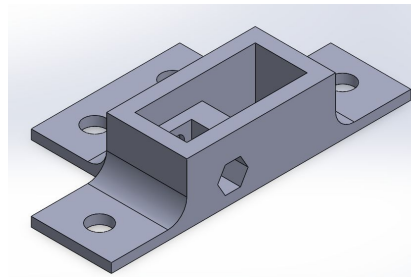
# Base Redesign

## Design Components

- Base Plate
- Motor cases

## Design considerations:

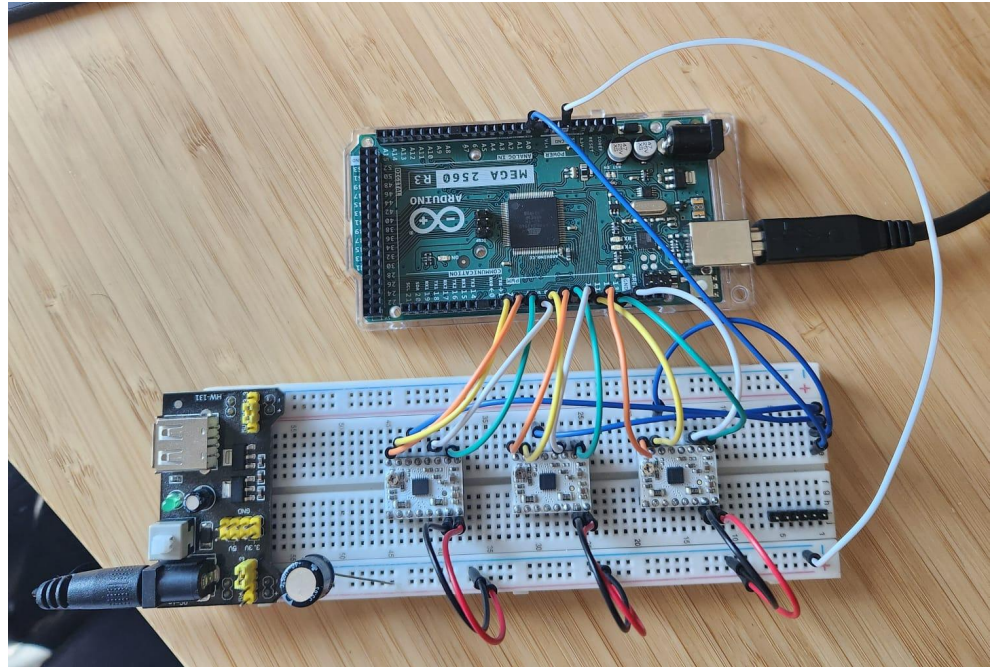
- Motors must be placed 120 degrees away from each other
- Distance of ball and socket joint from mounting ring to the actuator tip
- Separate components to keep the motors in place and screwed onto the base plate



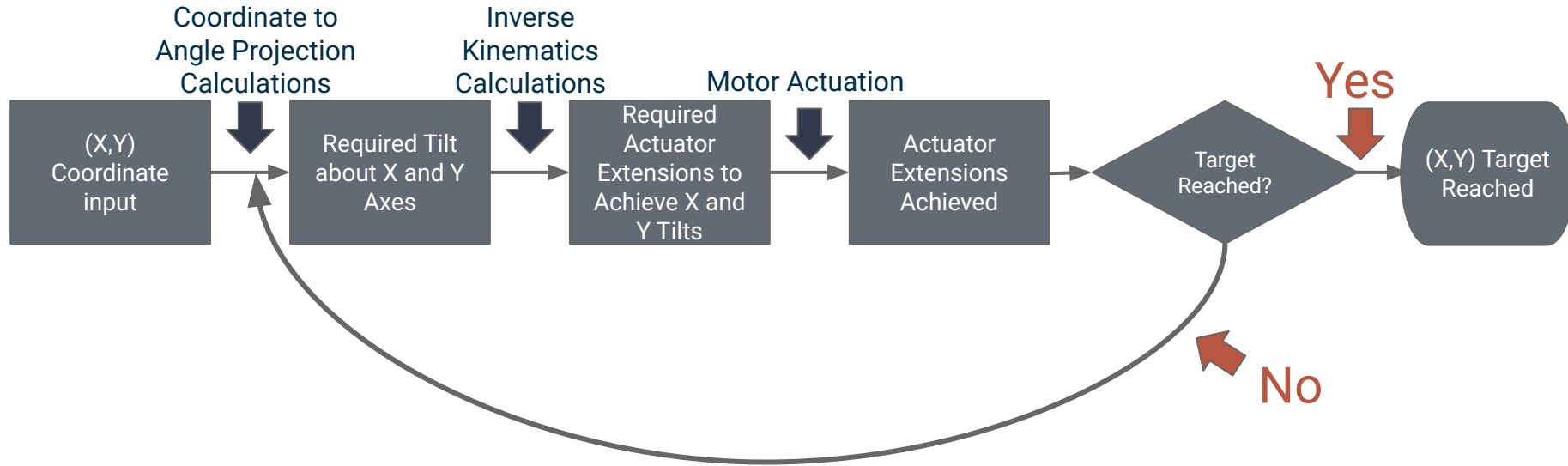
# Electronics



- Arduino Mega
- Motor Driver - DRV8834
- Low Voltage Power Source
- Solder pins to motor boards
- Solder pins to motor connector
- Accelerometer



# Coding Outline





# Coding Outline

| File                          | Functionality                                         |
|-------------------------------|-------------------------------------------------------|
| <i>main.ino</i>               | Orchestrates the full pipeline and handles user input |
| <i>config.h</i>               | Stores all system constants and parameters            |
| <i>ik.h</i>                   | Converts tilt angles to actuator extensions           |
| <i>actuation_sketch.h</i>     | Commands motors to target positions                   |
| <i>drv8834_accelstepper.h</i> | Low-level stepper motor driver control                |
| <i>paths.h</i>                | Defines movement paths and sequences                  |
| <i>orientation_check.h</i>    | Reads accelerometer and retries if off target         |

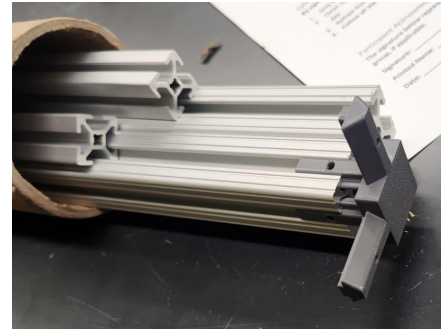
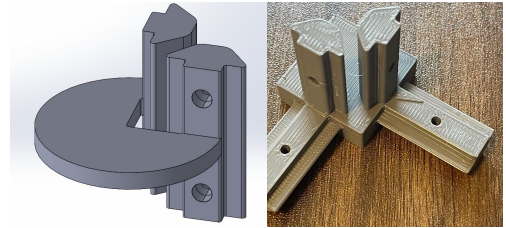
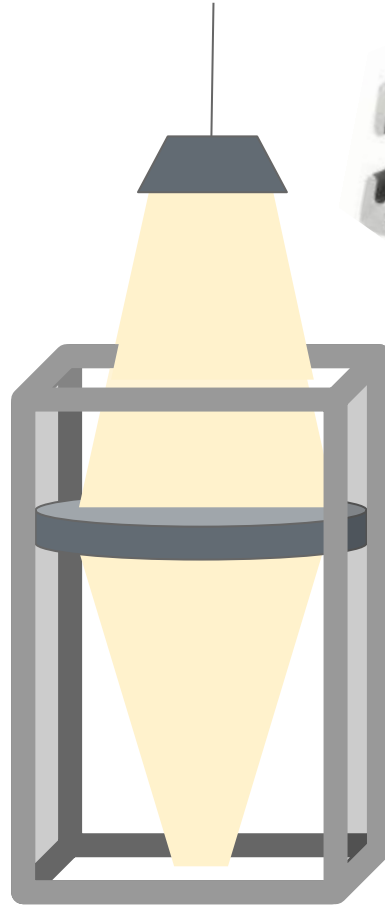


# Testing Apparatus Redesign

T-slot Aluminum extrusion frames, allow us to adjust the height of the lens to exactly the focal length.

Special components were designed and 3D printed to:

- Support and adjust the base height
- Hold the T-slots together at frame corners



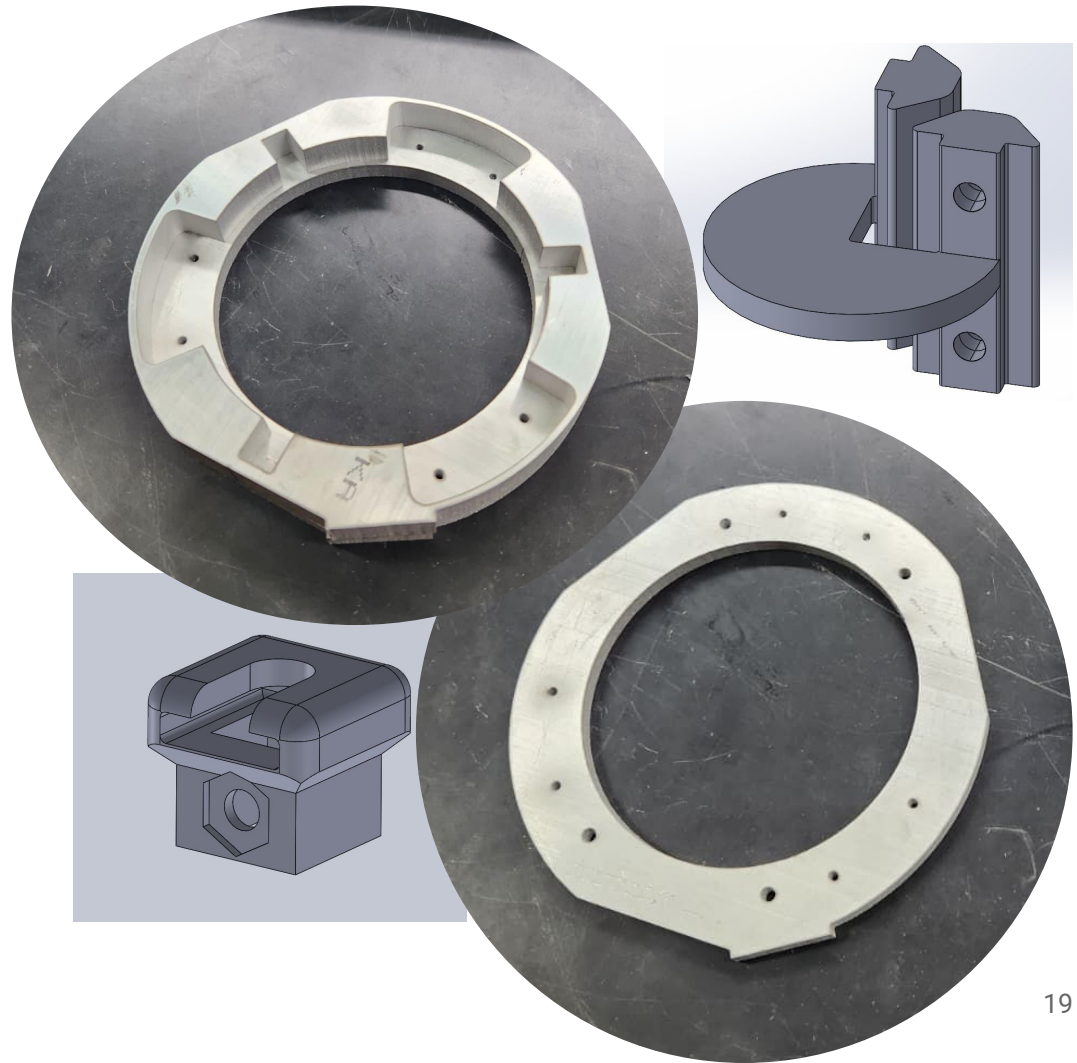
# Summary & Conclusion

In this presentation:

- Project motivation and description:
  - Mechanical system to direct concentrated light
- Discussed our design process from the previous semester
- Discussed the redesign of subsystems
  - Mounting ring, Base, and Testing apparatus
- Discussed the reselection and purchase of material components
  - Lens
  - Motor, Accelerometer, and Arduino control board
  - T-slot Frames and Ball and socket joints

# Next Steps

- Complete physical prototype
- Complete the Testing apparatus
- Test the design and code
- Iterate until completion



# End of Presentation!

Thank you,

The floor is open to questions